

NAVNEET KUMAR SINGH

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SUMMARY

Software Engineer with **4+ years of experience** architecting, developing, and deploying high-throughput microservices, RESTful APIs, and distributed cloud applications. Deep technical expertise in **Java and Spring Boot**, specializing in decomposing legacy monoliths, optimizing database performance, and building secure, event-driven architectures using **Kafka, Docker, and AWS**. Proven track record of delivering enterprise-grade software across the full software development lifecycle (SDLC)—partnering cross-functionally with Product, QA, and DevOps teams in Agile environments to build resilient systems with **99.99% operational uptime**.

SKILLS

Core Programming Languages: Java (Java 8/11/17), Python, JavaScript, TypeScript, SQL, HTML5/CSS3

Enterprise Java & Microservices: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, Spring Cloud, Spring Batch, Servlets, JSP, EJB, JDBC, Spring JDBC Template, RESTful APIs, Microservices Architecture, Node.js, Express.js, FastAPI

Frontend Technologies: React.js, Redux, WebSockets, STOMP, Socket.IO, JSON

Databases & Caching: PostgreSQL (pgvector), MySQL, MongoDB, Redis (Distributed Caching & Locking)

Cloud & DevOps: AWS (EC2, S3), Docker, Docker Compose, Nginx, GitHub Actions CI/CD, Maven, Gradle, Linux/Unix, PM2, systemd

Testing & Quality Assurance: JUnit 5, Mockito, Spock Framework (Groovy), SonarQube, Postman, Swagger / OpenAPI, TDD

Messaging & Architecture: Apache Kafka, RabbitMQ, EDA, Design Patterns, JWT, OAuth2, RBAC, Git, JIRA, Confluence

AI & Applied ML Integration: OpenAI API (GPT-4o), RAG Pipelines, Vector Search, Spring AI, LangChain, Embeddings Integration

EXPERIENCE

MetLife, United States | *Software Engineer*

January 2026 – Present

- Engineered high-throughput microservices using **Java17**, **Spring-Boot 3**, and **JPA**, decomposing core legacy policy administration monoliths into domain-driven REST APIs processing **1.2M+ daily insurance** claim events with 99.99% operational uptime.
- Architected an event-driven data streaming pipeline using **Apache Kafka** and **Spring Cloud Stream**, reducing real-time claim notification latency from 450ms to 85ms across 12 downstream policy microservices.
- Integrated an enterprise **RAG microservice** using **Spring AI**, **Python** (FastAPI), **OpenAI API** (GPT-4o), and **PostgreSQL** (pgvector), automating policy underwriting risk evaluations to cut manual document review time by 55%.
- Optimized database access across multi-terabyte policy databases by implementing **Redis** distributed caching and refactoring **Hibernate ORM** mappings, cutting p99 claims API response latency by 40%.
- Constructed** automated integration test suites and **GitHub Actions CI/CD** quality gates using **JUnit 5**, **Mockito**, **Spock**, and **SonarQube**, deploying containerized services to **AWS (EC2/S3)** while cutting regression defects by 30%.

Infosys Limited (client: Caterpillar Inc.), India | *Senior System Engineer*

September 2021 – August 2024

- Architected** and executed the modernization of an enterprise dealer service management system for **Caterpillar Inc.** (Fortune 50), collaborating in an Agile team of 6 to decompose a legacy monolith into 8 domain-driven microservices, reducing Mean Time to Repair (MTTR) by 40%.
- Optimized** backend REST APIs by identifying and eliminating N+1 query patterns in **Java** and **Spring Boot**, refactoring **Spring Data JPA / Hibernate** mappings to reduce response times by 20% and cut database roundtrips by 35%.
- Engineered** an automated **SLA breach** tracking and notification engine using **Spring Scheduler** across 300+ global dealerships, accelerating order processing workflows and improving pickup rates by 20%.
- Integrated Spring Security** and JWT-based **Role-Based Access Control** (RBAC) across **microservice** endpoints while authoring automated unit/integration test suites using **JUnit** and **Mockito** to maintain 90%+ code coverage.

Space Infolab, India | *Software Engineer*

July 2020 – August 2021

- Engineered asynchronous messaging workers using **Java 11**, **Spring MVC**, and **RabbitMQ** to offload high-volume reporting jobs from core application threads, boosting system request throughput by 35%.
- Built** full-stack administrative dashboards using **Node.js**, **Express.js**, and **React.js** deployed on **AWS (EC2)**, enabling internal operations teams to monitor client provisioning and system audit logs in real time.
- Refactored batch data processing pipelines using **Spring Batch** and **Spring JDBC Template**, optimizing multi-million row night-time ETL runs and cutting batch processing duration from 4 hours to 90 minutes.

EDUCATION

MS in Computer Science, *DePaul University, United States*

September 2024 – June 2026

BS in Computer Science, *Maharishi Markandeshwar University, Mullana, Ambala*

August 2017 – May 2021

PROJECTS

Real-Time Order Engine & Distributed Ledger Service | *Java 17, Spring Boot 3, Spring Cloud, gRPC, Redis, PostgreSQL, Docker*

- Architected a low-latency order execution engine using **Java 17**, **Spring Boot 3**, and **gRPC**, enabling sub-20ms inter-service communication across distributed microservices during peak trading windows.
- Engineered write-behind multi-level caching using **Redis** and **Spring Data JPA**, achieving 99.9% cache hit ratios and reducing database write bottlenecks across 500K+ daily transactions.

AI Document Search & RAG Vector Engine | *Java 17, Spring Boot 3, OpenAI API, PostgreSQL, Apache PDFBox, Docker*

- Engineered** a Retrieval-Augmented Generation (RAG) backend engine using **Java 17**, **Spring Boot 3**, and **pgvector** that parsed unstructured PDF documentation into 1536-dimensional vector embeddings, executing cosine similarity search via **OpenAI API** to cut document discovery time by 60%.